

Indian Institute of Technology Dharwad

CS201L: Artificial Intelligence Laboratory

Lab 4: Data Preprocessing: Outlier Detection, Correlation Analysis (Pearson and Spearman Correlation), Attribute Normalization and PCA

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1 Problem Statement

An energy efficiency dataset is provided in a CSV file (`energy_efficiency.csv`). The dataset contains various attributes that describe the geometric, architectural, and design characteristics of residential buildings. These features influence the thermal behavior of buildings and are used to analyze and predict their energy performance, specifically the heating and cooling requirements. The attributes included in the dataset are:

Table 1: Description of Variables in the Energy Efficiency Dataset

Variable	Name	Description
X1	Relative Compactness	Measure of building shape compactness affecting energy efficiency.
X2	Surface Area	Total external surface area influencing heat exchange.
X3	Wall Area	Total exterior wall area affecting thermal transfer.
X4	Roof Area	Roof surface area affecting heat gain and loss.
X5	Overall Height	Height of the building impacting thermal behavior.
X6	Orientation	Building direction relative to cardinal points.
X7	Glazing Area	Total window area influencing light and heat transfer.
X8	Glazing Area Distribution	Distribution of glazing across building façades.
Y1	Heating Load	Energy required to heat the building.
Y2	Cooling Load	Energy required to cool the building.

Write a Python program to perform the following tasks on the following attributes: X1, X2, X3, X4, X5, X6, X7, X8, Y1:

1.1 Outlier Detection

1. Read the dataset into a pandas dataframe. Plot the boxplot of the attributes. Observe the number of outliers present in each attribute and list their values. Outliers are defined as the data values that do not satisfy the following condition: $(Q_1 - 1.5 \times IQR) < x < (Q_3 + 1.5 \times IQR)$ where $IQR = Q_3 - Q_1$, Q_1 = Lower Quartile and Q_3 = Upper Quartile.
2. Replace these outliers with the median of the attribute. After outlier correction, save it as **energy_efficiency_corrected.csv** file since it is required in the next task. Plot the boxplot again and observe the difference.

1.2 Correlation

Degree of Correlation	Positive Correlation	Negative Correlation
Perfect Correlation	+1.0	-1.0
Very Strong Correlation	(0.5, 1.0)	(-1.0, -0.5)
Strong Correlation	(0.3, 0.5]	[-0.5, -0.3)
Moderate Correlation	(0.1, 0.3]	[-0.3, -0.1)
Weak Correlation	(0.0, 0.1]	[-0.1, 0.0)
Zero / No Correlation (Uncorrelated)	0.0	0.0

Table 2: Degree of Correlation

1. Load the outlier-corrected dataset **energy_efficiency_corrected.csv** and compute the Pearson and Spearman correlation coefficients for every pair of attributes in the dataset.
2. Visualize these correlation coefficients as a correlation matrix. (Use the heatmap plot to visualize it). Observe the correlation coefficients and the degree of correlation (shown in Table 2), and print the relation for each attribute pair based on the degree of correlation.
3. Let's assume Y2 as the target attribute. Now, find and visualize (use bar plot) the Pearson's and Spearman's correlation coefficients of the target attribute with all attributes for the dataset. Further, show the scatter plots between the target attribute and all other attributes and analyze them with their respective correlation coefficient.

1.3 Normalization and Standardization

1. Load the outlier-corrected dataset **energy_efficiency_corrected.csv** and observe the range of the values in each attribute. Find the minimum and maximum values in each attribute.

2. Load the outlier-corrected dataset `energy_efficiency_corrected.csv` and perform the Min-Max normalization of this data using **MinMaxScaler** to have the range of values between 0-1. After performing the operation save this file as `energy_efficiency_normalisation.csv`
3. Load the outlier-corrected dataset `energy_efficiency_corrected.csv` and perform Min-Max normalization to have the range of values between 10-30. After performing the operation save this file as `energy_efficiency_normalisation_30.csv`
4. Load the outlier-corrected dataset `energy_efficiency_corrected.csv` and find the mean and standard deviation of the attributes. Standardize each attribute using the **StandardScaler** command. Compare the mean and standard deviations before and after the standardization. After performing the operation save this file as `energy_efficiency_standardisation.csv`

For **MinMaxScaler**: <https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.MinMaxScaler.html>

For **StandardScaler**: <https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.StandardScaler.html>

1.4 Principal Component Analysis (PCA)

1. Load the outlier-corrected dataset `energy_efficiency_corrected.csv` and perform Principal Component Analysis (PCA) on the specified attributes. Fit the PCA model to the data and transform it accordingly. Analyze the resulting eigenvalues and eigenvectors by completing the following tasks:
 - (a) Compute and display the eigenvalues and their corresponding eigenvectors of the covariance matrix.
 - (b) Plot the eigenvalues in descending order.
 - (c) Plot the cumulative explained variance and clearly indicate the point corresponding to **95% variance retention**.
 - (d) Transform the dataset using:
 - i. all principal components and save the transformed data as `energy_efficiency_pca_all.csv`.
 - ii. the selected principal components that retain **95%** of the total variance, and save the transformed data as `energy_efficiency_pca_select.csv`. Plot the transformed data using scatter plot.
 - (e) Compute the covariance matrix of the transformed data obtained using:
 - i. all principal components, and
 - ii. the selected principal components based on **95% variance retention**.

For **Covariance**: <https://numpy.org/doc/2.1/reference/generated/numpy.cov.html>

For **Eigen values and vectors**: <https://numpy.org/devdocs/reference/generated/numpy.linalg.eig.html>

For **cumsum**: <https://numpy.org/devdocs/reference/generated/numpy.cumsum.html>

For **PCA**: <https://scikit-learn.org/stable/modules/generated/sklearn.decomposition.PCA.html>